一、Login screen

In the RSPAS Platform login screen (as shown in Figure 1), the user enters the registered email address and password, and clicks on the "General User Login" button or the "Administrator Login" button to enter the RSPAS Platform. If the user enters an incorrect email address or password, an error box will be displayed.

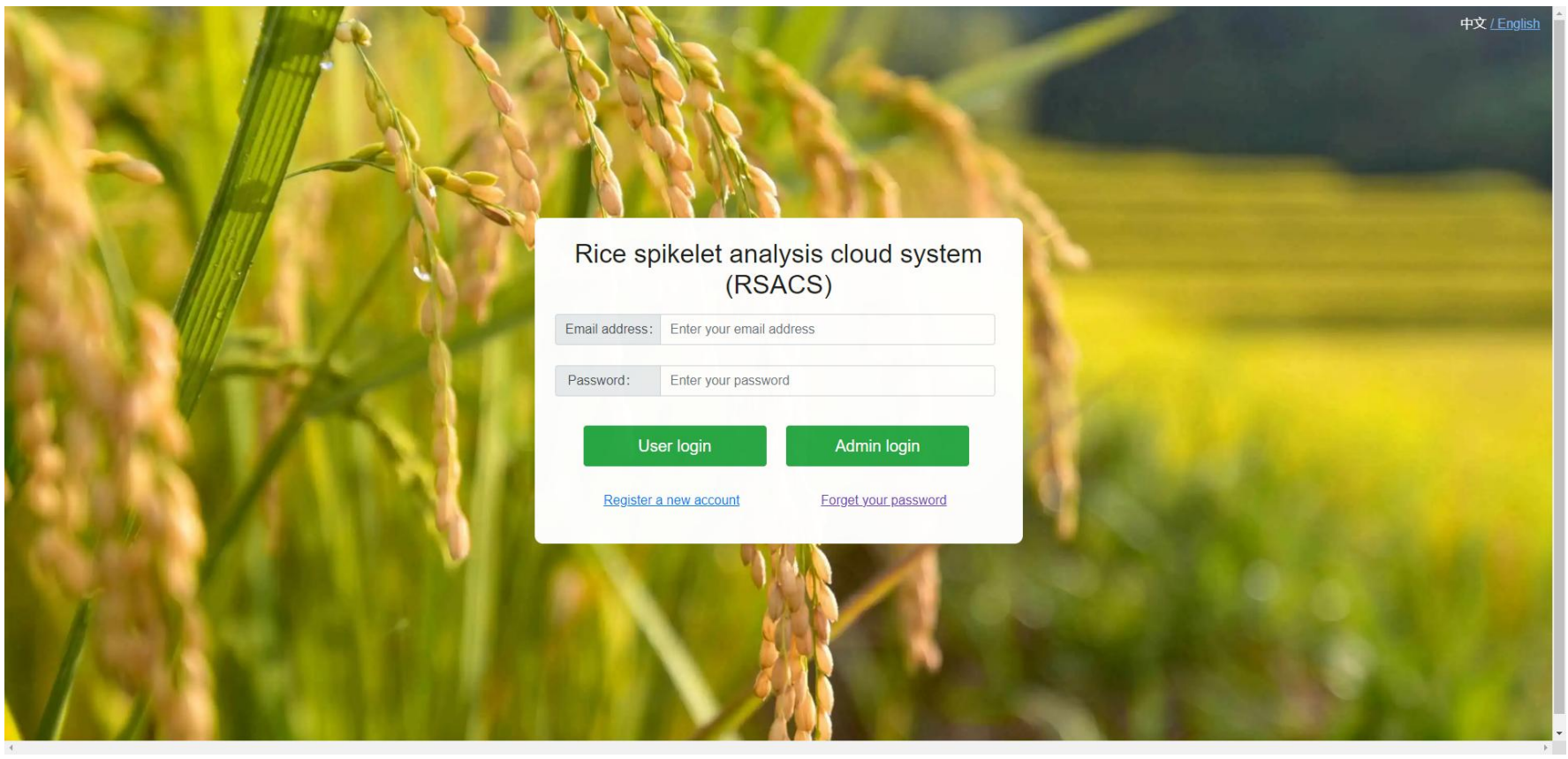


Figure1 Login screen of the RSPAS Platform

二、Registration screen

Click "No account? Click Register" to enter the RSPAS Platform registration screen (shown in Figure 2), according to the prompted information in the box, enter the required user name, password and email address, click "Register" button to complete the registration.



Figure2 Registration screen of the RSPAS Platform

三、The RSPAS Platform screen

3.1 The spikelet counting screen

When a user successfully logs in to the RSPAS, the initialisation interface for spikelet counting (shown in Figure 3), which is divided into five parts, is the default:

1、Upload an image

2、Choose appropriate an AI model

3、Counting results

4、Import a rice spikelet classification model

5、Display complete rice spikelets

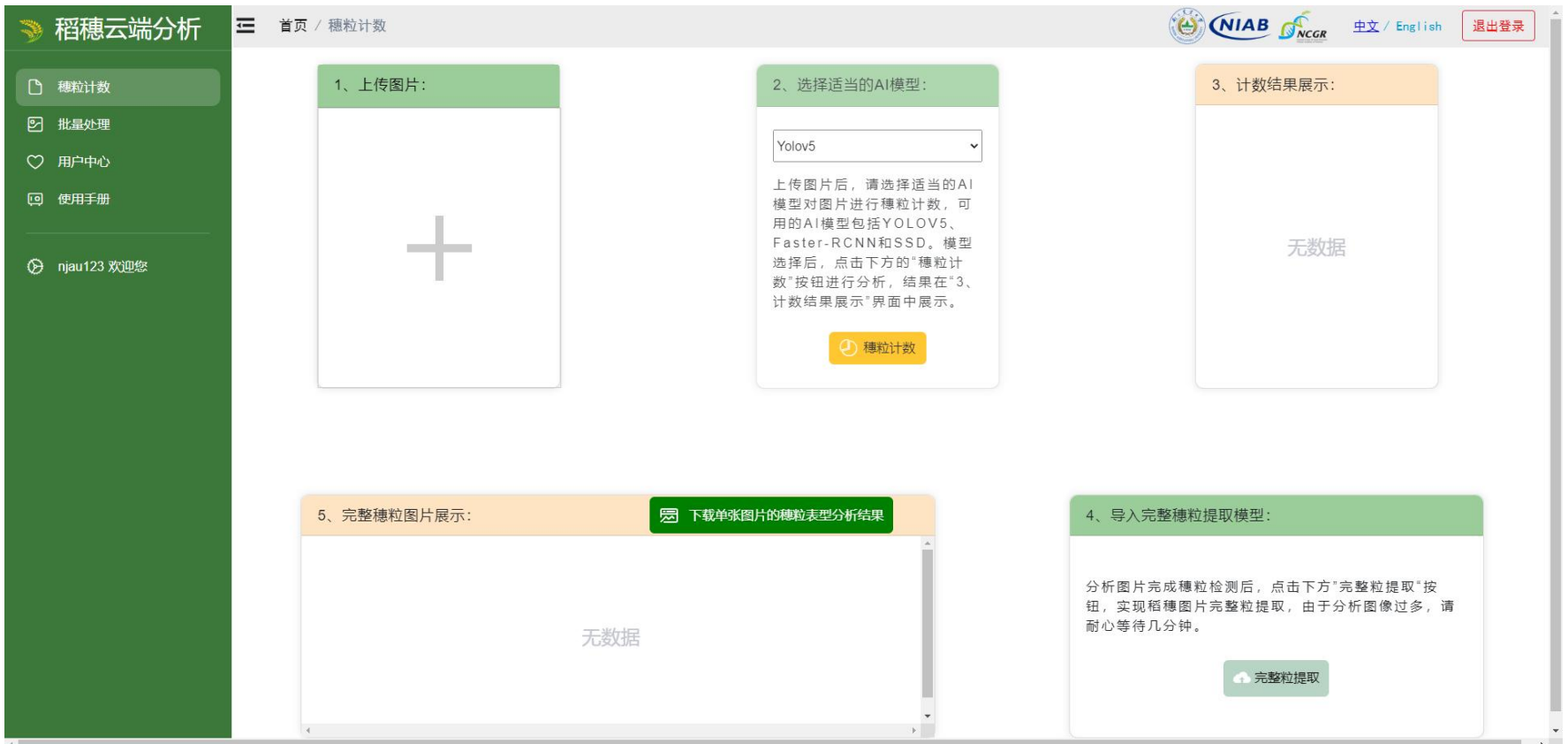


Figure3 The RSPAS Platform screen

3.1.1 Upload an image section

In the Upload an Image section, click on the + button and after selecting the rice image to be processed, the words Upload Image will change to a button to select a file.

3.1.2 Choose an AI model section

The user must choose the appropriate AI model from the three drop-down boxes, categorised into three target detection models, YOLOv5, Faster R-CNN and SSD, in the Select Appropriate AI Model section. Once the selection has been made, click on the Count spikelets button.

3.1.3 Counting results section

In the counting results display section, after the user clicks on the Count spikelets button, the user needs to wait first, and after the detection is complete, the complete picture of the detection and the counting result will be displayed on the page.

3.1.4 Import a rice spikelet classification model section

In the Import a rice spikelet classificarion model section, after the user clicks on the Complete spikelets extraction button, the complete spikelets are classified using the VGG model.

3.1.5 Dsiplay complete rice spikelets section

In the complete spikelet images display section, the result of the processing will be displayed in the complete spikelet images display box to show the classified complete spikelet images and the number of complete spikelets.

3.1.6 Download the results of the phenotypic analysis of spikelets in a single image

After the user has finished the execution in the above order, the page shown in Figure 4 below will be obtained, and the download of the processing results can be achieved by clicking on the button to download the results of the phenotypic analysis of spikelets in a single image, and the interface shown in Figure 5 will be obtained.

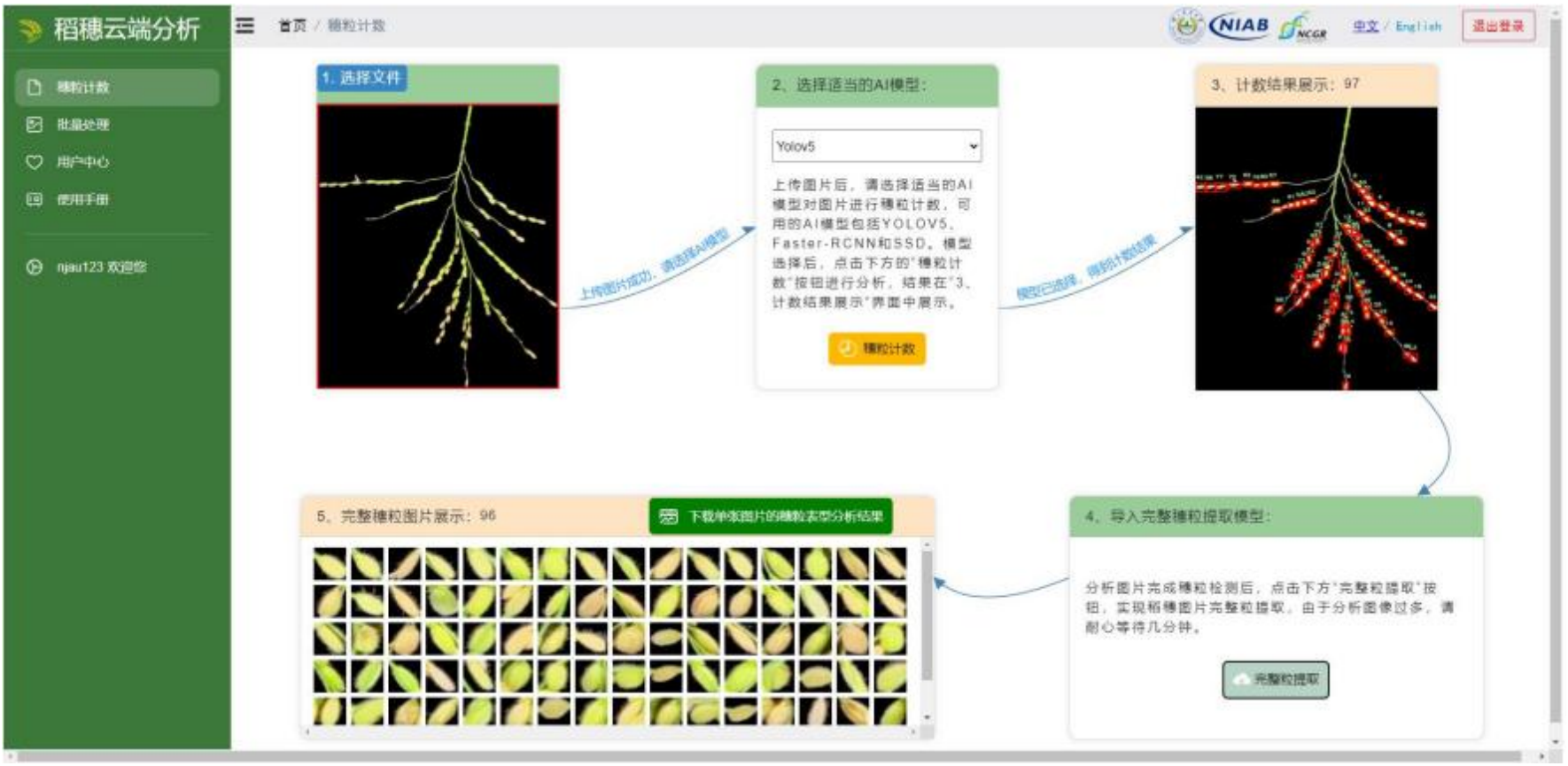
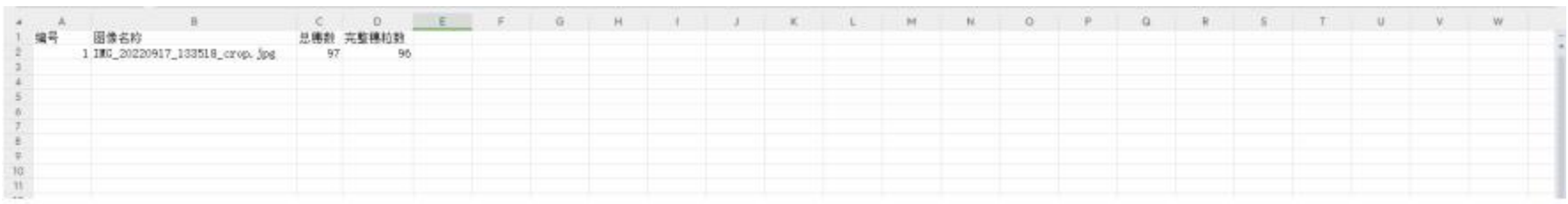


Figure4 Screen after spikelet counting treatment



编号	图像名称	总穗数	完整穗粒数
1	IMG_20220917_133518_crop_.jpg	97	96

Figure5 Download a single image of the spikelet phenotyping results interface

3.2 Batch processing screen

When the user clicks on Batch Processing in the side navigation bar,the user will enter the initialisation interface of Batch Processing (shown in Figure 6), which is divided into the following four sections:

1、Batch processing of images

2、Choose appropriate an AI model

3、Batch analysis

4、Results display

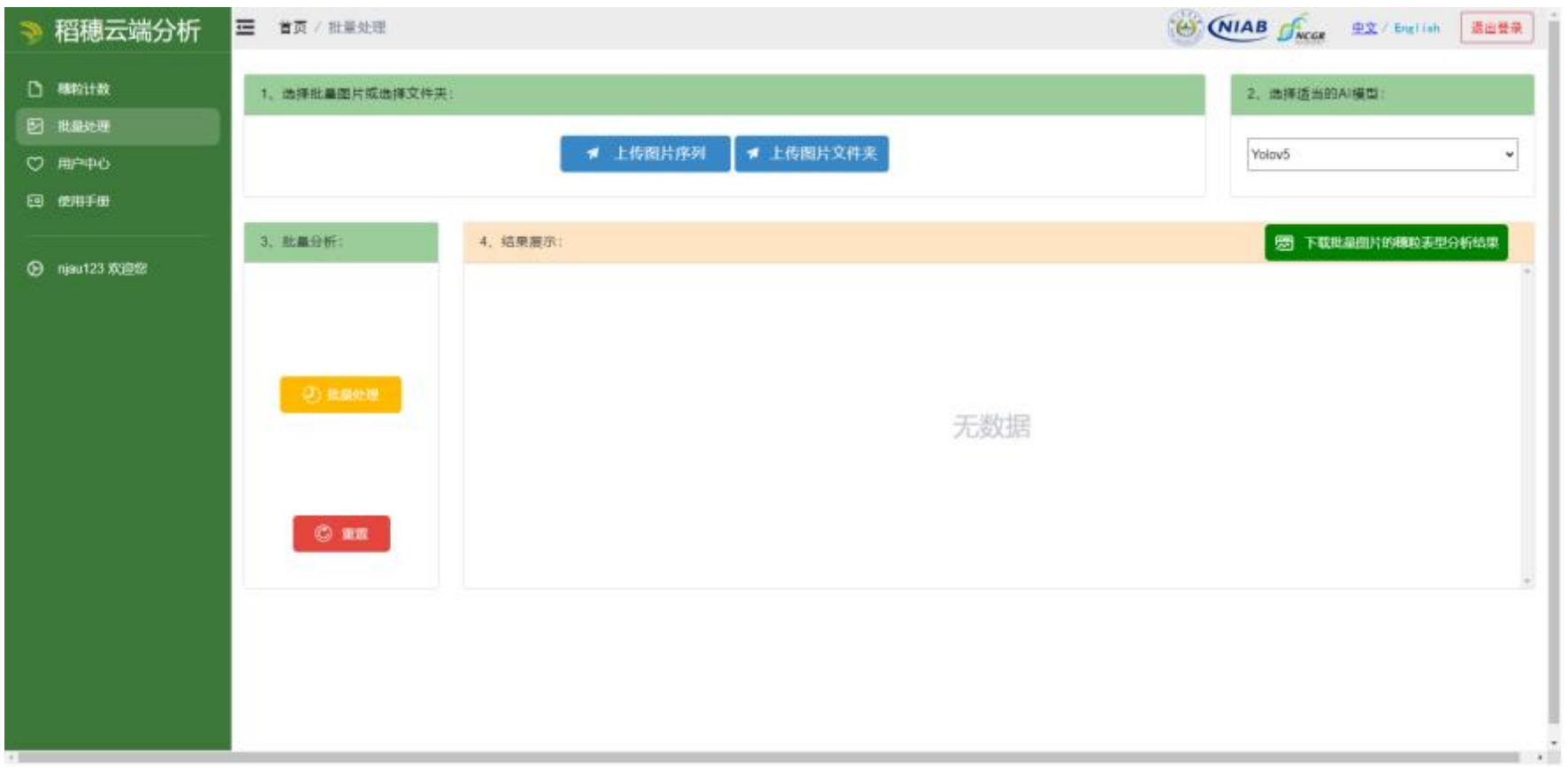


Figure6 Batch Processing Initialisation Screen

3.2.1 Batch uploading of images section

In the batch uploading of images section, the user can choose both batch upload images and upload folders for batch processing.

3.2.2 Choose appropriate an AI model

In the Choose Appropriate AI Model section, the user need to choose the appropriate AI model from the 3 drop-down boxes, which are generally categorised into three target detection models, namely YOLOv5, Faster R-CNN and SSD. Once the selection is complete, click on the Batch Processing button.

3.2.3 Results display

In the results display section, after the user clicks on the batch processing button, the user needs to wait, firstly the spikelets counting will be done according to the selected AI model, after the counting is done, the complete spikelets will be classified according to the detected spikelets, and finally the generated table will be displayed in the no-data section, which will show the identification, the name of the picture, the initial picture, the processed picture, the number of grains per spikelet, "Number of grains in a complete spikelet, as well as the picture of the complete spikelets.

3.2.4 Download the results of the phenotypic analysis of spikelets in batch images

After the user completes the above sequence, the interface shown in Figure 7 below is obtained. Clicking on the button to download the results of the phenotypic analysis of spikelets in batch images will enable the download of the processing results, resulting in the interface shown in Figure 8.

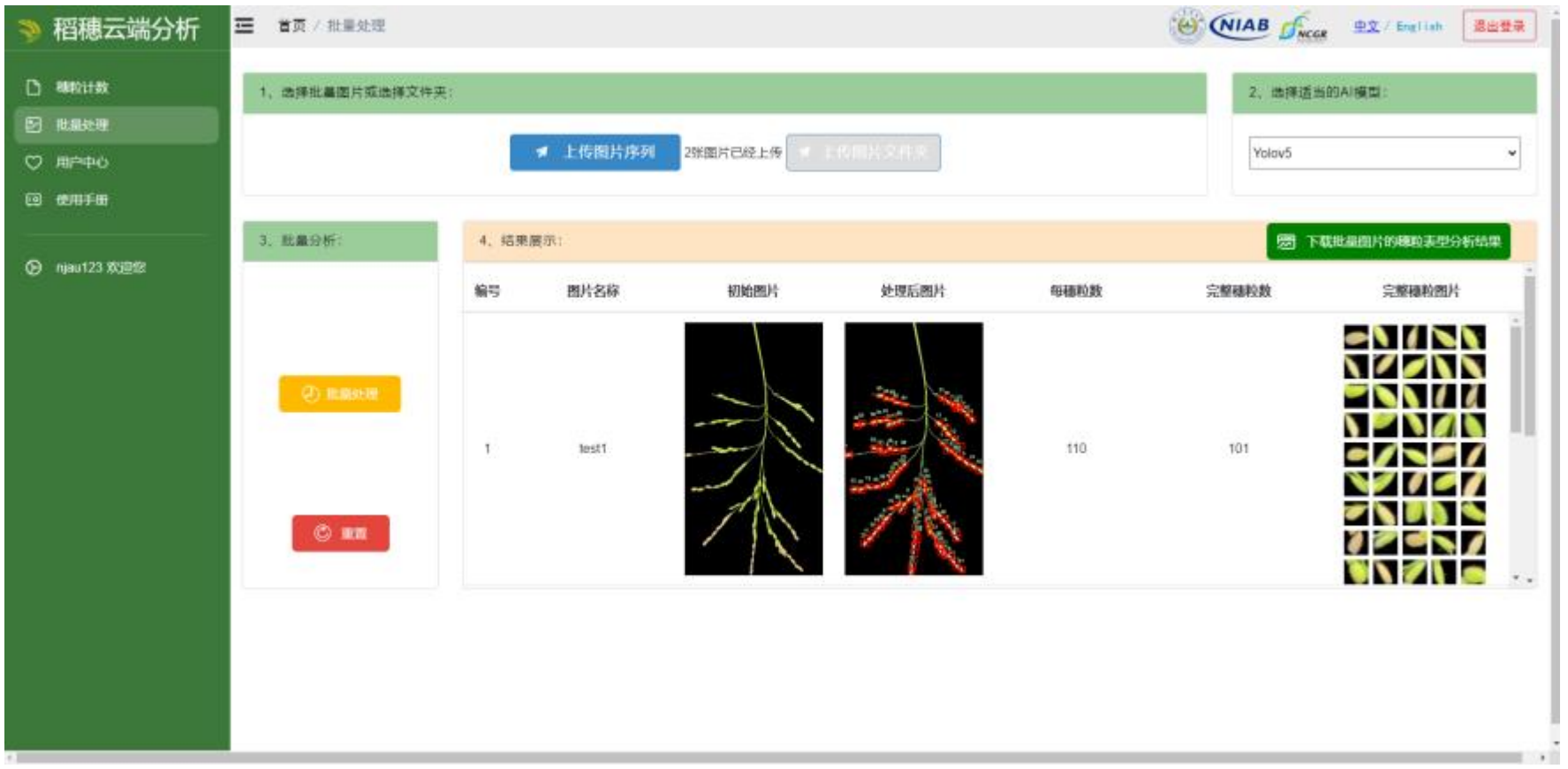


Figure7 Screen after Batch processing treatment

	A	B	C	D	E	F	G	H	I	J	I
1	编号	图片名称	总穗数	完整穗粒数							
2	1	1.jpg	110	101							
3	2	2.jpg	181	175							
4											
5											

Figure8 Download Batch images of the spikelet phenotyping results interface

3.3 User settings screen

If the user is an administrator, clicking on the User Centre in the side navigation bar will take you to the initialization interface of the User Centre as shown in Figure 9, which is mainly responsible for editing and deleting users. If you are an ordinary user, click Personal Centre in the side navigation bar to enter the initial interface of Personal Centre as shown in Figure 10, which is mainly responsible for editing the information of the currently logged-in users.



Figure9 User Centre initialisation screen

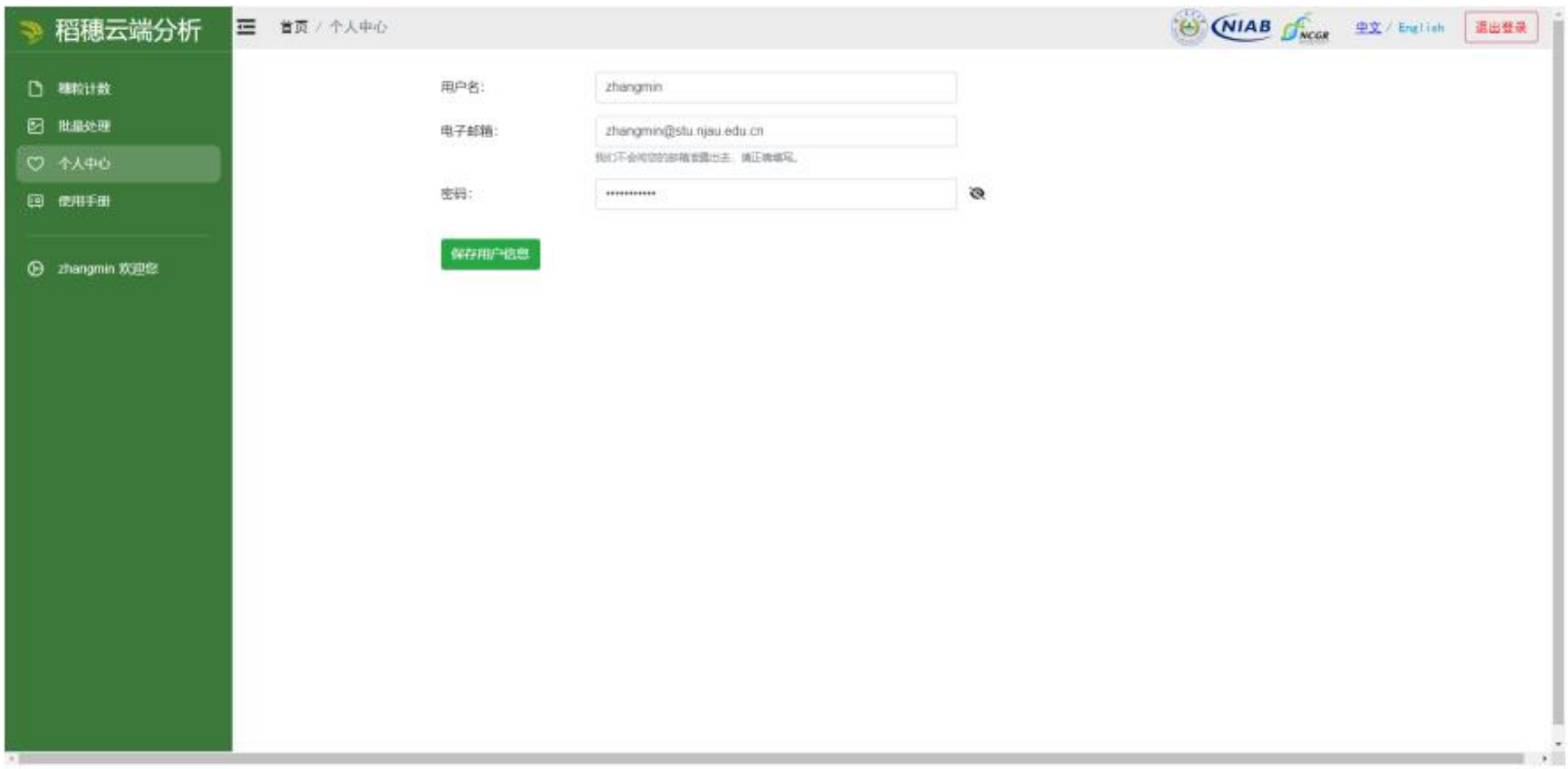


Figure10 Initialisation screen of the Individual centre

The administrator can perform editing operations on any user in the table by clicking the Edit button to edit the user's information, as shown in Figure 11 below.

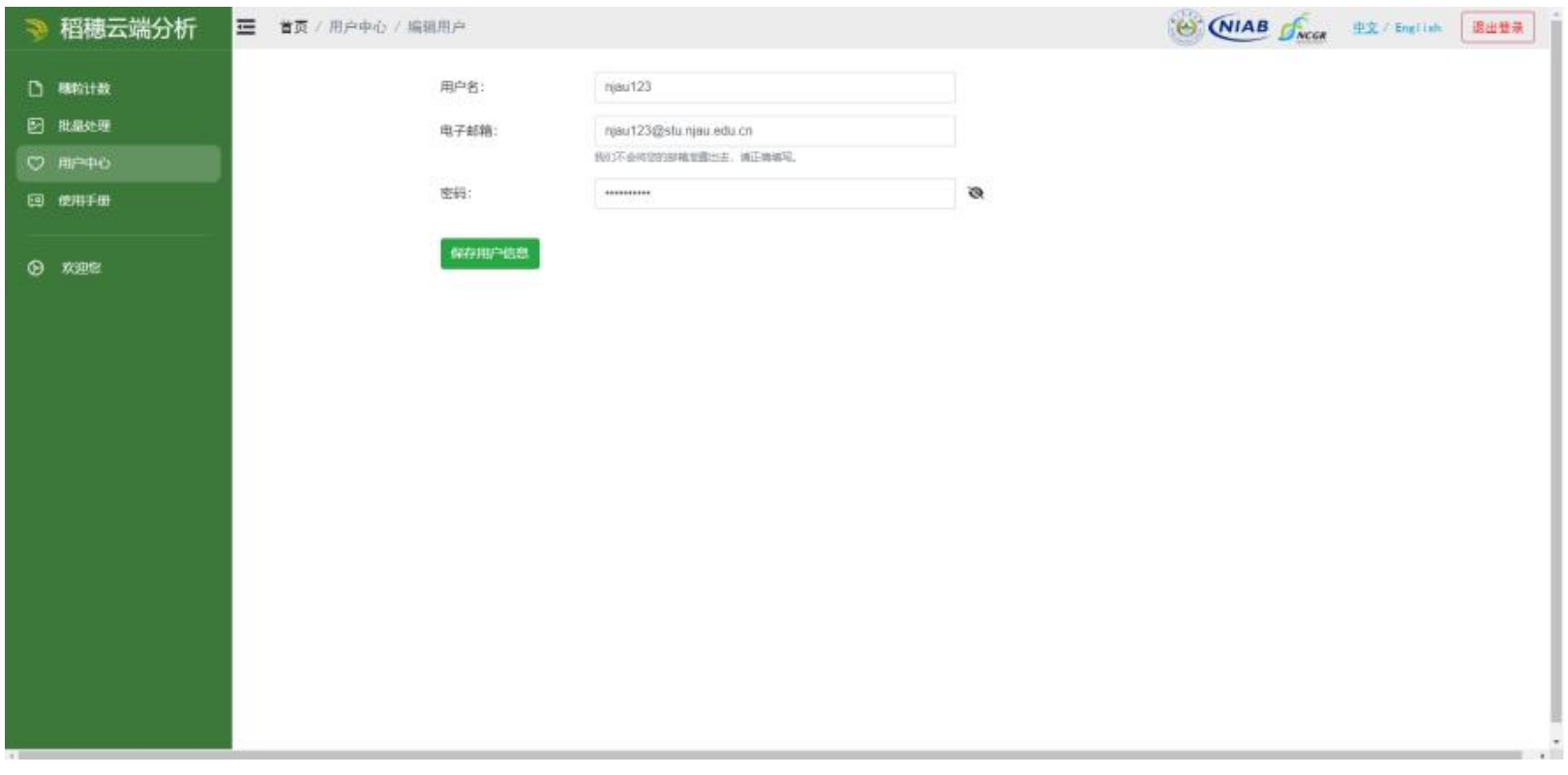


Figure11 Edit user screen

3.4 User manual section

The user can enter the interface of the user manual by clicking on the user manual in the left navigation bar, which can help the user to operate the RSPAS, and the initialisation interface is shown in Figure 12 below.



Figure12 Initialisation screen of the User manual